

# Therapeutic approaches to protein-misfolding diseases

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**Several sporadic and genetic diseases are caused by protein misfolding. These include cystic fibrosis and other devastating diseases of childhood as well as Alzheimer's, Parkinson's and other debilitating maladies of the elderly. A unified view of the molecular and cellular pathogenesis of these conditions has led to the search for chemical chaperones that can slow, arrest or revert disease progression. Molecules are now emerging that link our biophysical insights with our therapeutic aspirations.**

**H**ow and whether a protein folds *in vivo* are influenced primarily by its amino-acid sequence and the cellular environment surrounding the polypeptide chain<sup>1</sup>. Protein folding in the cell occurs either in the cytoplasm or within the secretory pathway (see the review in this issue by Dobson, page 884). ATP-dependent chaperones in the cytoplasm and the lumen of the endoplasmic reticulum (ER) collaborate to give a polypeptide chain several opportunities to fold. If folding is unsuccessful, the polypeptide is directed to the proteasome — a large multisubunit protease — for degradation (see the review in this issue by Goldberg, page 895). Ensuring accuracy in protein folding is crucial for maintaining proper cellular function.

Several diseases are caused by a loss of protein function because the misfolded proteins have been degraded by the proteasome. Cystic fibrosis and  $\alpha$ 1-antitrypsin ( $\alpha$ 1-AT) deficiency are well-known heritable diseases that fall into this category<sup>2–4</sup>. Other misfolding diseases are caused by conformational changes coupled to the aggregation of misfolded proteins outside the cell, beyond the influence of intracellular quality-control systems. The major representatives of these disorders are the amyloidoses, in which protein aggregation in the extracellular space is associated with the appearance of a toxic function<sup>5,6</sup>.

Proteins that use the secretory pathway for folding and trafficking are over-represented in protein-misfolding diseases. These proteins fold in the ER and then proceed through the secretory pathway to their final destination, where the environment can be very different. For example, lysosomal enzymes have to fold at neutral pH to leave the ER, yet their sequences are optimized for folding and stability at pH 5 (lysosomal pH). So mutations in lysosomal proteins can adversely alter folding in the ER, leading to degradation by the quality-control machinery. Ironically, many of the variants that cause heritable lysosomal storage diseases would be able to fold into functional enzymes in the lysosome, if only they could get there<sup>7,8</sup>. In contrast, amyloidogenic proteins seem to fold efficiently in the ER but get into trouble in their destination environments, where they misfold and misassemble, a process known as amyloidogenesis. The thermodynamic principles behind protein folding are discussed in Box 1.

Therapeutic intervention can take the form of drugs that reduce the production of a certain protein, drugs that increase the rate of clearance of misfolded/aggregated pro-

teins, and drugs that increase the native-state stability or increase the kinetic barrier to misfolding aggregation (see Box 1).

In this review, we focus on small molecules, antibodies and dominant-negative protein subunits that increase the stability of the native state and/or increase the activation barrier associated with misfolding because these approaches are innovative and likely to be efficacious.

## Small-molecule chaperones and treatment of disease

Many wild-type (WT) proteins fold inefficiently in the ER, even with the aid of the heat shock protein 40/70/90 (hsp40, hsp70 and hsp90) chaperone system that facilitates the refolding or retrotranslocation of misfolded proteins back into the cytoplasm, where they are degraded by the proteasome. Mutations further reduce folding efficiency by altering the energetics of proper folding. However, small-molecule ligands that bind to the native state of these mutant proteins can stabilize the native state or destabilize the transition state (see Box 1) to compensate for the influence of the mutation (Fig. 1).

Bouvier and colleagues<sup>9</sup> have shown that selective V2-receptor antagonists (for example, SR121463A and VPA985) can lead to a 15-fold increase in expression of functional V2 vasopressin receptor at the cell surface, rescuing these receptors from proteasome degradation. Antagonists that were not cell permeable were ineffective, showing the importance of intercepting the misfolded protein in the cell and most likely in the ER. Chemical chaperones are small molecules that bind to a protein, stabilize the folded state and thereby reduce protein misfolding. The cell-permeable chemical chaperones were able to functionally complement more than six V2 receptor variants, some of which are associated with nephrogenic diabetes insipidus. These investigators also used chemical chaperones to complement the misfolding tendencies of variants of the  $\delta$ -opioid receptor<sup>10</sup>.

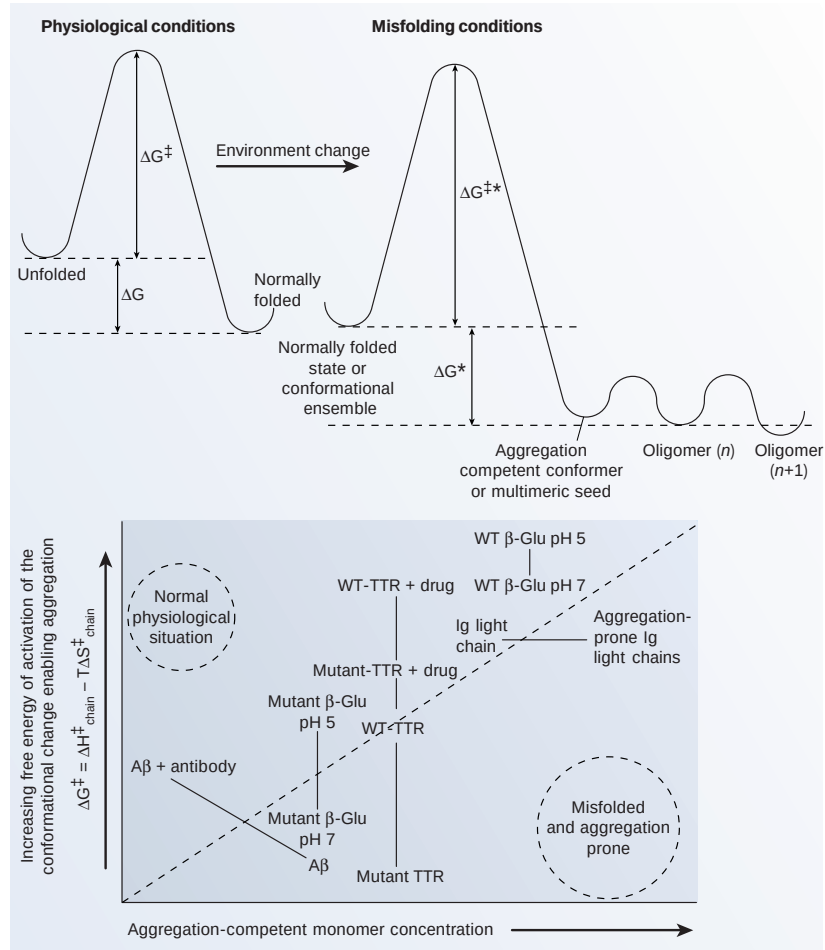
Conn and colleagues<sup>11</sup> have demonstrated that it is possible to use indoles, quinolones and erythromycin macrolides to complement the secretion-deficient loss-of-function variants of the gonadotropin-releasing hormone receptor (GnRHR) in transiently transfected COS-7 cells. ER-permeable agonists and antagonists with a high affinity for WT GnRHR were most efficient at chaperoning several, but not all, GnRHR variants out of the ER, and enabling their trafficking to the plasma membrane where they function. In an analogous approach, defects in rhodopsin folding in the

# Box 1

## The thermodynamics of protein folding

In discussing therapeutic approaches to diseases of protein misfolding, it is useful to consider the kinetic and thermodynamic underpinnings (see figure). Most proteins easily find their native state because the free energy of activation ( $\Delta G^\ddagger$ , which dictates rate) separating the unfolded and transition states is surmountable, and the folded state is substantially more stable than the unfolded state (the difference in free energy,  $\Delta G$ , dictates the relative populations of folded and unfolded proteins). For aggregation-prone proteins, it has become clear that the misfolded oligomeric state can be more stable than the functional three-dimensional structure under certain conditions, including those associated with a high-denaturation stress<sup>47,48</sup>. Fortunately, the kinetic barrier,  $\Delta G^\ddagger$ , separating the native and misfolded states is usually insurmountable on a biologically relevant time frame, and, in most cases, intrinsic clearance mechanisms operate more efficiently than the misfolding process<sup>19,49</sup>. Because the misfolding process involves multimerization of individual chains,  $\Delta G^\ddagger$  has two components, one relating to the conformation of the monomer and a second relating to multimerization that is dependent on concentration. The asterisk indicates that this  $\Delta G^\ddagger$  is the barrier between the monomeric conformer and a multimer that can oligomerize efficiently.

The free-energy diagram in the top left panel of the figure indicates to what extent a given sequence under native conditions prefers a normally folded conformation. The difference in free energy ( $\Delta G$ ) between the native state and the unfolded ensemble of conformations (unfolded state) dictates the relative population of each state. The free energy of activation ( $\Delta G^\ddagger$ ) governs how fast one conformation will convert to another. Under denaturing conditions, a misfolded aggregation-competent conformation can be preferred to the normally folded native state as a consequence of the interplay between sequence and environment<sup>1</sup>, as shown in the top right panel of the figure. For example, the environment surrounding the polypeptide can change as a result of intracellular or extracellular alterations.  $\Delta G$  and  $\Delta G^\ddagger$  have implications analogous to those described above. The bottom panel of the figure illustrates the interplay between sequence, monomer concentration and the presence of drug or



antibody in altering the free energy of activation,  $\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$ , for amyloidogenesis. Situations below the diagonal permit amyloidogenesis, whereas those found above the diagonal resist misassembly.

After this energetic analysis, it is possible to consider therapeutics in three classes: interventions that decrease production; small molecules, antibodies or dominant-negative protein subunits that increase native state stability and/or the kinetic barrier to misfolding/aggregation; and approaches that increase clearance of the misfolded monomer, soluble multimer or insoluble amyloid.

disease retinitis pigmentosa were corrected by using retinal-based ligands<sup>12</sup>, as were human *ether-a-gogo-related* gene (*HERG*) mutations by channel blockers<sup>13</sup>.

The most prevalent lysosomal storage disease, Gaucher's disease, is caused by the inefficient folding and trafficking of certain variants of  $\beta$ -glucocerebrosidase ( $\beta$ -Glu). The most prominent mutation (N370S) and other  $\beta$ -Glu mutations were thought to alter active-site residues. However, it is now becoming clear that several of these mutations interfere with folding in the ER and/or trafficking. The intracellular activity of misfolded variants such as N370S can be partly restored by administering cell-permeable  $\beta$ -Glu inhibitors that serve as chemical chaperones. These inhibitors seem to provide a template that complements the active site of  $\beta$ -Glu, stabilizing the native state in the ER, allowing  $\beta$ -Glu to leave the ER and be trafficked to the lysosome. The high concentration of the glycolipid substrate in the

lysosomes of  $\beta$ -Glu-deficient cells can competitively displace the inhibitor, allowing the enzyme to function. Kelly and colleagues<sup>8</sup> have shown that a subset of N-alkylated deoxynojirimycins and simpler six-membered ring N-heterocycles increase the activity of  $\beta$ -Glu within human cells<sup>8</sup>; the best compounds discovered so far double  $\beta$ -Glu activity. This should be sufficient to ameliorate the pathology of Gaucher's disease and could offer a convenient alternative to intravenous enzyme replacement therapy; however, relevant transgenic animal models to test this approach are not yet available.

Fan *et al.*<sup>14</sup> have shown that it is possible to use the small molecule 1-deoxy-galactonojirimycin, a potent competitive inhibitor of  $\alpha$ -galactosidase A ( $\alpha$ -Gal A), to rescue the misfolding and mistrafficking associated with certain Fabry-disease-causing mutations. These chemical chaperones facilitate the proper folding and trafficking of  $\alpha$ -Gal A variants to the lysosome, increasing enzyme activity

seven- to eightfold in cells when used at sub-inhibitory intracellular concentrations. Biophysical experiments suggest that these chemical chaperones stabilize the enzyme. Oral administration of 1-deoxygalactonojirimycin to transgenic mice with Fabry disease expressing a variant of  $\alpha$ -Gal A increased the enzyme activity significantly in several organs<sup>14</sup>. Chaperoning of  $\alpha$ -Gal A can also be accomplished by galactose infusions in humans<sup>15</sup>.

Other loss-of-function diseases, including cystic fibrosis, might be corrected by chemical chaperones<sup>2,3</sup>. Cystic fibrosis caused by the  $\Delta F508$  mutation in the CF channel protein results from chaperone-mediated retention of this mutant in the ER and subsequent degradation of this channel by the proteasome. Investigators at Vertex and a group at the University of California, San Francisco have recently identified small molecules that bind to and stabilize the  $\Delta F508$  CF channel in the ER, enabling the channel to exit from the ER and to be trafficked to the plasma membrane, where it exhibits sufficient activity to partly ameliorate the disease<sup>16</sup>.

$\alpha 1$ -AT is a member of the serpin family of protease inhibitors and is secreted by liver cells to control the proteolytic activity of a variety of circulating enzymes. In  $\alpha 1$ -AT deficiency, a misfolded but otherwise functional mutant protein is retained in the ER of liver cells. The failure to secrete  $\alpha 1$ -AT into the circulation allows neutrophil elastase to degrade lung parenchyma, causing emphysema. Perlmutter and colleagues<sup>17</sup> have shown that osmolytes such as 4-phenylbutyric acid and glycerol increase the secretion efficiency of  $\alpha 1$ -AT in cell lines and in transgenic animals. Although these compounds do not bind to  $\alpha 1$ -AT specifically, they increase the secretion efficiency of variants of  $\alpha 1$ -AT that are normally retained in the ER. 4-Phenylbutyric acid and glycerol accomplish this without influencing the secretion efficiency of other proteins or by decreasing proteasome activity. We speculate that these osmolytes upregulate the chaperone system in cells and in transgenic mice expressing variants of  $\alpha 1$ -AT. This would explain why misfolded variants of  $\alpha 1$ -AT, but not other WT proteins, are secreted more efficiently. We suspect that this is not a pure osmolytic effect in cells because the most potent osmolyte, trimethylamine N-oxide, does not increase secretion efficiency. This is reminiscent of the impact of guanidinium chloride, a potent denaturant, on curing the prion  $\text{Psi}^+$  phenotype in yeast, a result attributable to chaperone changes.

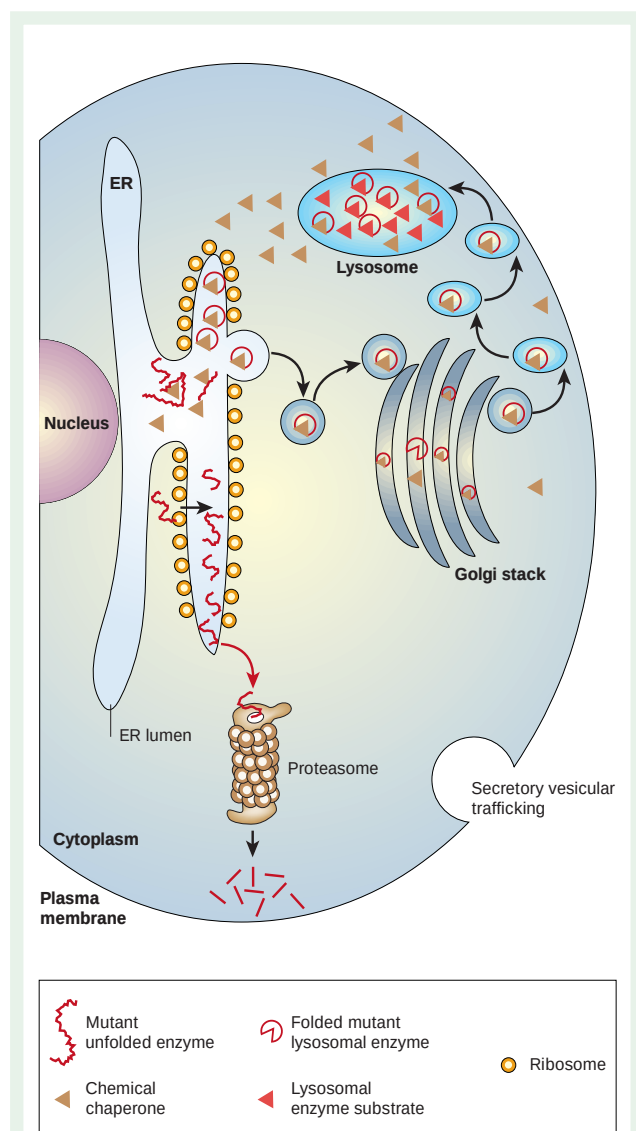
### Stabilizing proteins that misfold after secretion

The misfolding and misassembly that occur within the ER are largely controlled by the interplay between the chaperone and proteasome systems. Misfolding and misassembly occurring later in the secretory pathway or outside the cell, after secretion or membrane attachment, are more difficult to correct. Extracellular amyloid and related aggregates can be degraded by immune-system pathways including catabolism by macrophages. Unfortunately, aggregate clearance is often slowed down by disease and ageing, leading to the accumulation of misfolded and misassembled proteins thought to be responsible for diseases such as Alzheimer's and the familial amyloidoses.

### Transthyretin amyloidosis

Stabilization of the native state is one approach to prevent post-secretory misfolding that has been studied extensively in the case of transthyretin (TTR) amyloidogenesis. TTR is secreted into the plasma as a homotetramer with 127 amino-acid subunits, and is the backup transporter for thyroxine and the primary transporter of holo-retinol-binding protein. Tetramer dissociation to a misfolded monomer is typically rate-limiting for TTR misassembly, and leads to the formation of non-fibrillar and amyloid aggregates. All of the TTR mutations associated with familial disease studied so far decrease tetramer stability and, in the most severe cases, increase the speed of dissociation and thus of amyloidogenesis<sup>18</sup>. Interallelic trans-suppression of TTR amyloidogenesis has been observed in a compound heterozygous Portuguese family, who express the disease-associated mutation V30M but do not develop familial amyloid

polyneuropathy<sup>19</sup>. These individuals make TTR tetramers composed of T119M and/or V30M subunits. The incorporation of T119M subunits imparts kinetic stability by increasing the tetramer-dissociation barrier, slowing the rate of TTR-amyloid fibril formation<sup>19</sup>. This result suggests that interallelic and intra-allelic trans-suppression of misfolding could become very powerful therapeutic strategies for the prevention of misfolding diseases. It might be difficult to use suppressor proteins delivered by exogenous supplementation owing to the slow kinetics of subunit exchange and the possible antigenicity of suppressor variants<sup>20</sup>. Similar approaches have been suggested to prevent p53 misfolding in a variety of cancers<sup>21</sup>. Gene therapy, once practical, should be very useful for the implementation of trans-suppression.



**Figure 1** Folding of mutant proteins destined to be lysosomal enzymes. Cell-permeable chemical chaperones (dark red triangles) act as a template for variant lysosomal enzymes as they fold in the ER. Some of these lysosomal enzymes would otherwise misfold and be retained in the ER and ultimately be degraded by the proteasome. Chemical chaperones facilitate folding in the ER lumen, which is necessary for proper trafficking to the lysosome where a high concentration of substrate (red triangles) exists to compete with the chemical chaperones that bind to the active site. Although the chemical chaperones are crucial for folding at pH 7 (for example, in the ER), the variant lysosomal enzymes might not need the chaperones when they arrive in the lysosome (where the pH is about 5) because their sequences are stably folded at this pH.



Evidence that trans-suppression ameliorates TTR amyloid disease provides both an incentive and the biological rationale to search for small molecules that stabilize the native state and/or destabilize the transition state associated with TTR amyloidogenesis<sup>22</sup>. Screening and structure-based design have identified six distinct classes of small molecules that bind with high affinity to the unoccupied thyroxine-binding sites within TTR. These small molecules stabilize the native state, preventing amyloidogenesis and denaturation at low pH and in a wide variety of chaotropes and organic solvents for extended periods (for example, weeks)<sup>19</sup>. When ligand binding stabilizes the native state by more than 3 kcal mol<sup>-1</sup> (about 12.5 kJ mol<sup>-1</sup>), the concentration of the misfolded amyloidogenic monomer decreases by more than 100-fold (assuming that the barriers associated with misfolding are readily surmountable). Decreasing the concentration of the aggregation-prone species is important because amyloidogenesis is a high-order kinetic process; decreasing the concentration modestly can therefore markedly influence the aggregation rate<sup>23</sup>. Stabilization of the native state by a ligand also has kinetic ramifications when the ligand selectively stabilizes the native state over the transition state<sup>19</sup>. TTR tetramer stabilization by ligand binding often increases the  $\Delta G^\ddagger$  of dissociation (see Box 1), markedly slowing the typical rate-limiting step of TTR amyloidogenesis. The mechanistic similarities between trans-suppression and native-state small-molecule stabilization imply that altering denaturation energetics with ligands will probably be effective in humans.

#### Prion disease

The agreement between the various mouse models of prion disease and their human and bovine counterparts provides a prototypic setting for the exploration of therapeutic approaches to diseases of misfolding. These therapeutic efforts can be divided into two broad classes, serendipitous and mechanistic. A variety of compounds including anthracyclines, porphyrins and diazo dyes block prion replication when administered with PrP<sup>Sc</sup>, the aggregated infectious form of the prion protein, into animal models<sup>24,25</sup>. Unfortunately, this is not a clinically relevant model for therapeutic intervention, because subclinical disease exists for months in mice and years in humans. Efforts to identify more potent analogues of these molecules have been disappointing; however, quinacrine might prove to be a different case. As quinacrine is an approved pharmaceutical, clinical trials are being carried out to test the usefulness of this molecule in patients with Creutzfeldt–Jakob disease. Medicinal chemistry efforts are underway to identify more potent agents. For example, some bisacridines are 10–20-fold more potent than quinacrine. Animal studies are underway to characterize the efficacy of bisacridines in the treatment of prion disease. Recently, Zahn and colleagues<sup>26</sup> have used NMR spectroscopy and chemical-shift titration to show that quinacrine binds specifically to PrP<sup>C</sup>, the normally folded cellular isoform of the prion protein. This observation could accelerate drug-development efforts.

Studies of heterozygous and homozygous mice with PrP knocked out show that decreasing or eliminating PrP<sup>C</sup> production will slow or prevent prion infection. Similarly, antibody studies suggest that more efficient clearance of PrP could slow or cure prion disease<sup>27–29</sup>. These results complement similar efforts with Alzheimer's disease<sup>30</sup>. As with other aggregation-prone proteins, the tendency for PrP<sup>C</sup> to misfold can be altered by proteins that bind to PrP<sup>C</sup>. In prion disease, PrP<sup>Sc</sup> clearly induces misfolding, so antibodies or small molecules that antagonize this interaction should prove useful. There is also genetic evidence that other host molecules facilitate PrP<sup>Sc</sup> replication<sup>31</sup>. Dominant-negative variants of PrP<sup>C</sup> slow or stop PrP<sup>Sc</sup> replication in cell culture and in transgenic animals. Small molecules designed to mimic this dominant-negative effect can block PrP<sup>Sc</sup> replication in cell culture as well.

#### Multiple myeloma

Immunoglobulin light-chain amyloidoses are particularly challenging from a therapeutic perspective. Multiple myeloma (plasma-cell

cancer) or plasma-cell dyscrasia overproduce an antibody light-chain clone that is prone to aggregation. Light-chain aggregation often results in progressive organ failure, making it difficult to use autologous blood stem-cell transplantation to treat the cancerous component of this disease. Direct approaches to decrease light-chain production by the use of specific antibodies, antisense RNA or short interfering RNA are likely to be difficult owing to the patient-specific features of each plasma-cell dyscrasia<sup>32</sup>. However, it is possible that Velcade, a proteasome inhibitor that is effective in the treatment of multiple myeloma, could have a paradoxical palliative effect on light-chain aggregation. Plasma cells need an efficient proteasome system to manipulate antibodies, so proteasome inhibitors could disable or kill neoplastic cells faster than new cells arise, decreasing the manufacture of light chains.

#### Intervention in Alzheimer's disease

The risk of Alzheimer's disease is strongly influenced by two competitive ER processing pathways. The amyloid precursor protein (APP) has a single membrane-spanning helix that is processed by either  $\alpha$ -secretase (about 90%) or by the  $\beta$ - and  $\gamma$ -secretases (about 10%). The cleavage by  $\alpha$ -secretase produces two fragments that do not produce amyloids. The  $\beta$ -secretase and subsequent  $\gamma$ -secretase cleavages yield three fragments; the smallest of which is a family of 39–43-residue peptides referred to as A $\beta$ . These fragments have a tendency to assemble into a variety of aggregates, including amyloid fibrils. Generally speaking, the longer the A $\beta$  peptide, the more amyloidogenic it is. Its exact length distribution is controlled by the allosteric modulation of  $\gamma$ -secretase.

Inhibition of A $\beta$ -fibril formation might be a reasonable therapeutic strategy, because familial mutations that lead to an increase in A $\beta$  concentration or to its aggregation increase neuropathology<sup>33–35</sup>. Peptidomimetics, based on the peptide LVFFA from A $\beta$ , modified at the amino or carboxy terminus, and the all-D (right-handed) version and several retroinverso peptidomimetics, block both A $\beta$  seeding and growth. Unfortunately the pharmacological profile of these compounds is far from ideal<sup>33</sup>. Non-peptidic, aromatic-rich A $\beta$  aggregation inhibitors have also been reported by many pharmaceutical and biotechnology companies. These compounds typically inhibit the aggregation of A $\beta$  and several other amyloidogenic polypeptides. Their lack of specificity might be a problem; it now seems likely that mammalian melanocytes and related cell types make amyloid for functional purposes<sup>36</sup>. Moreover, recent evidence suggests that soluble, diffusible non-fibrillar A $\beta$  aggregates lead to pathology in Alzheimer's disease<sup>5,37–39</sup>. This calls into question the merits of the subset of these compounds that shift the distribution from fibrils towards soluble, diffusible non-fibrillar A $\beta$  aggregates.

Because A $\beta$  aggregation is highly dependent on monomer concentration (because of its high kinetic order) and on the distribution of A $\beta$  family members, substantial effort has gone into the development of inhibitors, activators and modulators of the secretases<sup>40,41</sup>. Decreasing the concentration of A $\beta$  below the critical concentration prevents aggregation<sup>42</sup>, and so it is not necessary to block all A $\beta$  production. Several  $\beta$ -secretase inhibitors are now available, and this seems to be a promising approach because mice lacking  $\beta$ -secretase do not have an overt phenotype. Considerable progress has also been made in developing  $\gamma$ -secretase inhibitors. However, the side-effect profile for these inhibitors must be considered carefully: mice lacking  $\gamma$ -secretase die *in utero* because this enzyme processes notch, a regulatory protein required for development. A therapeutic window for  $\gamma$ -secretase inhibitors will require substantial inhibition of A $\beta$  processing without the complete inhibition of notch processing. Golde and Koo<sup>43</sup> have shown that some non-steroidal anti-inflammatory drugs can influence the processing of APP to produce less A $\beta$  1–42; others have shown that cholesterol depletion can markedly decrease the overall amount of A $\beta$  produced.

When APP transgenic mice overexpressing A $\beta$  are actively immunized with A $\beta$  peptides, deposits that would ordinarily accumulate

are cleared<sup>44</sup>. Passive immunotherapy with intravenous A $\beta$  antibody also prevents the formation of A $\beta$  deposits in animals that would otherwise be deposit-laden. Unfortunately, active immunization of patients with Alzheimer's disease using A $\beta$  1–42 revealed antibodies crossing the blood–brain barrier causing meningoencephalitis-like inflammation in 4% of the trial participants, presumably originating from microglial activation<sup>45</sup>. This resulted in the discontinuation of the clinical trial. Cautious optimism still exists for passive immunotherapy in humans, because anti-A $\beta$  antibody infusion could be stopped if meningoencephalitis symptoms develop<sup>46</sup>.

## Future considerations

Although thermodynamics undoubtedly has a role in misfolding diseases, it is the kinetics of misfolding and misassembly that seems to have the dominant influence. New therapeutics are likely to come from medicinal chemistry efforts and clinical trial designs based on the considerations discussed here. We therefore expect that orally administered chemical chaperones that treat lysosomal storage diseases will replace intravenous enzyme replacement therapy as the standard treatment for these rare inherited conditions. Some drugs have already been identified that block the destruction of mutant integral membrane proteins exhibiting suboptimal folding in the ER. We believe that therapeutics based on similar approaches will soon enter clinical trials for patients with cystic fibrosis and  $\alpha$ 1-AT deficiency. For the amyloid-deposition diseases, small alterations in the rate of aggregation mediated by stabilization of the native state or depletion of the precursor could stabilize or improve patients with these diseases. Enhanced clearance of amyloidogenic aggregates by immunotherapy also seems promising. Given the challenges posed by unregulated immune responses after vaccination, passive immunotherapy is more likely to be available in the short term. More clinically palatable small-molecule approaches will follow with the more typical timeline for pharmaceutical discovery and development. The histological aggregates observed by Lewy, Pick and other noted pathologists more than a century ago have recently been identified as largely homogeneous deposits of misfolded proteins. A large and increasing number of diseases are now associated with the accelerated production and/or delayed clearance of protein aggregates. With the development of pharmacological approaches to these diseases, we are participating in a journey from the bedside to the bench and now back to the bedside. □

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